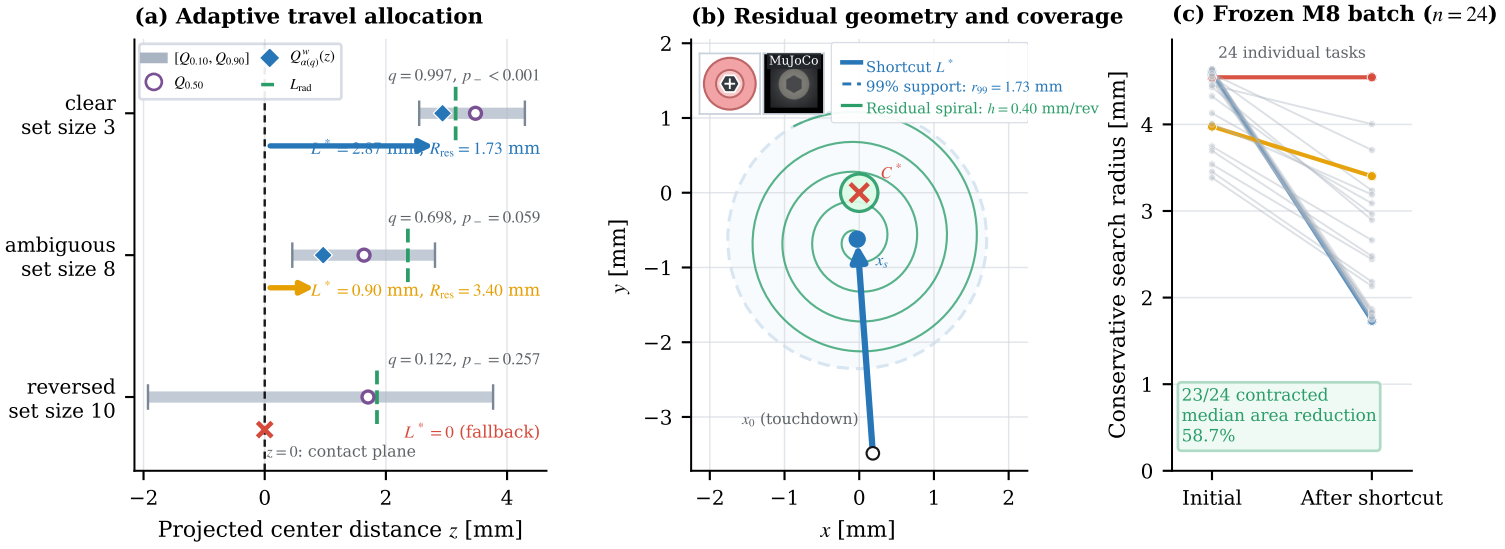

(a) Adaptive travel allocation
$[Q_{0.10}, Q_{0.90}]$
$Q_{0.50}$
$Q^{w}_{\alpha(q)}(z)$
L_{rad}
clear
set size 3
$q = 0.997$, $p_- < 0.001$
$L^* = 2.87$ mm, $R_{\mathrm{res}} = 1.73$ mm
ambiguous
set size 8
$q = 0.698$, $p_- = 0.059$
$L^* = 0.90$ mm, $R_{\mathrm{res}} = 3.40$ mm
reversed
set size 10
$q = 0.122$, $p_- = 0.257$
$L^* = 0$ (fallback)
$z = 0$: contact plane
−2
0
2
4
Projected center distance z [mm]
(b) Residual geometry and coverage
MuJoCo
Shortcut L^*
99% support: $r_{99} = 1.73$ mm
Residual spiral: $h = 0.40$ mm/rev
C^*
x_s
x_0 (touchdown)
y [mm]
x [mm]
(c) Frozen M8 batch ($n = 24$)
24 individual tasks
Conservative search radius [mm]
23/24 contracted
median area reduction
58.7%
Initial
After shortcut